

CampyTeq — Unified Multi-College Digital Ecosystem

Comprehensive System Architecture, Workflows & 12 Institutional Roles Specification
| Production v1.0

Document Version: 1.0.0 Production Spec	Backend: Django 5.1, DRF, PostgreSQL 16, Redis 7
Target Audience: Leadership, Tech Leads, DevOps, Auditing	Frontend: Next.js 14 App Router, TypeScript, Tailwind
Tenancy Model: Strict Row-Level Isolation (TenantModel)	Mobile Companion: React Native / Expo 51 (iOS & Android)
RBAC Coverage: 12 Granular Institutional Roles	Edge CV: RTSP Camera Ingestion & Journey Stepper

1. System Architecture & Engineering Principles

CampyTeq is architected as an enterprise-grade higher education digital operating system. It eliminates siloed institutional departmental tools by converging administrative governance, academics, examinations, biometric and lecture attendance, fees & invoicing, payroll, audience-scoped circulars, campus print kiosks, digital library circulation, edge CCTV safety tracking, and explainable AI student risk analysis under a unified multi-tenant umbrella.

- Multi-Tenant Core Isolation:** Every domain entity inherits from TenantModel, strictly binding rows to a college_id UUID. The active college context is resolved exclusively from the authenticated JWT session (request.user.college). TenantManager enforces database-level tenant isolation, preventing cross-tenant data leakage.
- Cross-Tenant Administrative Context Switching:** Users with the SUPER_ADMIN role are permitted to switch operational college contexts via the verified X-College-ID HTTP header without modifying base database credentials.
- Stateless API Architecture:** Built on Django REST Framework with SimpleJWT (access & refresh rotation), OpenAPI 3.0 auto-generation via drf-spectacular, and standardized JSON responses ({ success, message, data/errors }).
- Asynchronous & Edge Ingestion:** High-throughput edge computer vision telemetry from campus cameras stream into the tracking gateway without blocking synchronous student academic operations.
- Next.js 14 Single-Page Enterprise App:** Server components, client-side caching, reactive dark-mode interface, and 23 dedicated dashboard workflows tailored per authenticated role.

Note: Multi-Tenancy Invariant: Under no circumstances can client-supplied POST/PUT body parameters override the tenant identifier. The college context is strictly derived from the validated server-side token.

Layer	Component	Technology	Responsibility & Key Features
Edge / Gateway	Reverse Proxy & SSL	Nginx Alpine	SSL/TLS termination, Gzip compression, rate limiting (30 req/s general, 5 req/s token auth).
Web Frontend	Next.js App Router	Next.js 14, TypeScript, Tailwind	Interactive 23-route dashboard, 1-click demo role selector, role-scoped sidebar navigation, responsive modals.
Mobile App	Student Companion	React Native, Expo 51	Attendance radar, assignment submission status, fee invoice payments, print shop PIN retrieval, offline token handling.
Application Core	REST Backend	Django 5.1 DRF, Gunicorn	Tenant scoping, 12 RBAC role decorators, drf-spectacular OpenAPI docs, custom UUID User model.
Database	Relational DB	PostgreSQL 16 Alpine / SQLite fallback	Row-level logical separation, foreign keys, UUID primary keys, automated nightly SQL dump backup.
Cache & Queues	In-Memory Store	Redis 7 Alpine	JWT blacklist caching, session storage, Celery task broker, rate limiting state.

Edge Vision	CV Ingestion Stream	Python, OpenCV, RTSP Streamer	Camera frame parsing, biometric face recognition gateway, student sighting ingestion, zone telemetry.
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2. End-to-End Core Institutional Workflows

CampyTeq implements 10 deeply connected operational workflows that synchronize administration, academics, finances, and campus security.

Workflow 1: Academic Structural Hierarchy & Scheduling

- **Entity Hierarchy:** College → Academic Departments (headed by HOD) → Degree Courses (B.Tech, BCA, MCA) → Batches (e.g. 2023-2027) → Semesters → Subjects (Theory/Lab).
- **Faculty Allocation:** Teachers are mapped to specific subjects within their department. Cross-department elective assignments are supported.
- **Timetable Engine:** Weekly recurring schedules with room allocations, designated lecture slots, lab durations, and automatic conflict detection for faculty double-booking.

Workflow 2: Dual Attendance & Statutory Defaulter Radar

- **Lecture Attendance Marking:** Faculty select their active subject slot; system displays the enrolled student roster with 1-click 'Mark All Present' and individual toggle for 'Absent' or 'Late'.
- **Statutory 75% Defaulters Radar:** The analytics engine continuously calculates the percentage of attended classes per student. Students with < 75% attendance are automatically flagged on Principal, HOD, Mentor, and Student dashboards.
- **Faculty Biometric Timesheets:** Faculty log daily check-ins and check-outs, calculating net on-campus instructional and research hours.

Workflow 3: Examination Governance & Assignment Lifecycle

- **Exam Series:** Exam coordinators configure series (Midterms, End-Semester Finals, Practicals), publish schedules, and open grade entry windows.
- **Grade Entry & Confidential Publishing:** Faculty enter individual student scores; results remain confidential until approved by the Principal or Controller of Examinations.
- **Assignments Engine:** Faculty publish assignments with deadline timestamps, instructions, and attachments. Students submit coursework online; faculty provide grade marks and feedback.

Workflow 4: Financial Governance, Invoicing & Payroll Batches

- **Fee Structure Configuration:** Annual/semester tuition, laboratory, library, and examination fees assigned per course and batch.
- **Student Invoicing & Payment Reconciliation:** Invoices generated with itemized dues. Payment simulator records transaction IDs, updates status to 'PAID', and issues instant digital tax receipts.
- **Staff Payroll Batches:** Monthly salary batches computing base pay, DA/HRA allowances, tax deductions, and net salary. Confidentially dispatched to individual faculty profiles.

Workflow 5: Audience-Scoped Circulars & Leave Management

- **Targeted Circulars:** Administrative circulars scoped to target audiences ('ALL', 'STUDENTS', 'FACULTY', 'HOD', or specific departments) with file attachments and broadcast timestamps.
- **Multi-Tier Leave Workflow:** Students request leave to Mentors; Faculty request leave to HOD; HOD requests leave to Principal. Tracks leave balance, medical certificates, and approval audit trails.

Workflow 6: Campus Fast-Print Kiosk Queue

- **Job Submission:** Students and faculty upload documents (PDF/DOCX), specify copies, color vs. B/W, duplex, and binding options.
- **Instant Cost Estimation & 4-Digit Pickup PIN:** System calculates page costs, debits student wallet or flags payment, and issues an encrypted 4-digit PIN.

- **Kiosk Fulfillment:** Print staff enter the PIN at the campus kiosk terminal, releasing the document directly to connected network printers.

Workflow 7: Digital Library Catalog & Circulation Engine

- **Catalog Management:** ISBN, title, author, category, rack location, total copies, and active availability tracking.
- **Circulation:** Digital checkout and return timestamps. Automated calculation of overdue fines (■5/day beyond statutory loan duration).

Workflow 8: Edge Computer Vision Campus Tracking & Safety Radar

- **Camera Zone Registry:** Mapping of campus IP cameras across gates, library, academic blocks, cafeteria, and perimeter zones.
- **Edge Sighting Ingestion:** Edge camera service detects student presence and POSTs metadata timestamps and confidence scores to /api/v1/tracking/sightings/.
- **Student Journey Stepper:** Visual chronological timeline tracking student movement across campus zones.
- **Ethical Privacy Audit Log:** Authorized security searches require explicit logged justifications, preventing unmonitored surveillance abuse.

Workflow 9: AI Academic Early Warning Radar

- **Multi-Factor Explainable Risk Scoring:** Computes composite academic failure risk based on attendance (< 75%), internal test marks (< 40%), assignment submission timeliness, and fee overdue status.
- **Proactive Counseling Interventions:** Mentors receive algorithmic early warning notifications, scheduling one-on-one student counseling interventions before end-semester exam disqualification.

Workflow 10: Student Mobile Companion (React Native / Expo)

- **Cross-Platform Mobile App:** Provides students with real-time class schedules, live attendance percentages, upcoming assignment deadlines, fee invoice payment receipts, and print shop PINs.

3. Institutional Role Capabilities & Login Feature Breakdown

CampyTeq implements 12 strictly segregated RBAC roles. Each role receives a tailored dashboard, curated navigation tree, and dedicated business logic endpoints.

1. Super Admin — *superadmin@campyteq.io*

Primary Mandate: Global Multi-Tenant Platform Operator

- **Tenant Management:** Create, configure, suspend, and monitor multiple college instances across the SaaS network.
- **Cross-Tenant Switcher:** Switch context between colleges using verified X-College-ID headers to inspect live tenant states.
- **Global Audit & System Health:** System-wide error logs, API latency monitoring, and database health metrics.
- **User Administration:** Manage top-level administrators and tenant provisioning quotas.

2. Principal — *principal@apex.edu*

Primary Mandate: Chief Executive & Academic Head of Institution

- **Institutional Executive Overview:** Real-time institutional KPIs (total enrolled students, active faculty, overall attendance rate).
- **Campus Defaulter Radar:** College-wide monitoring of students below the mandatory 75% attendance threshold.
- **Departmental Oversight:** Review performance across Computer Science, Electronics, Mechanical, and Civil departments.
- **Exam Results Sign-Off:** Authorize and publish semester examination series results to public student portals.
- **Leave Final Escalation:** Approve multi-tier faculty and HOD leave requests.

3. Management / Trustee — *management@apex.edu*

Primary Mandate: Executive Board & Governance

- **Financial Velocity Dashboards:** Comprehensive fee collection velocity, outstanding student receivables, and budget allocations.
- **Payroll Expenditure Monitoring:** Monthly faculty payroll totals, institutional operational costs, and department expenditures.
- **Institutional Analytics:** Year-over-year enrollment trends, faculty student ratios, and strategic planning metrics.

4. Head of Department (HOD) — *hod.cs@apex.edu*

Primary Mandate: Departmental Academic & Staff Leader

- **Departmental Roster:** Complete oversight of all students, faculty members, and courses belonging to their department.
- **Subject & Faculty Mapping:** Assign faculty instructors to specific theory courses and laboratory batches.
- **Weekly Timetable Governance:** Configure and resolve lecture schedules and laboratory slot allocations.
- **Departmental Defaulter Alert:** Monitor student absenteeism specific to departmental classes.
- **Faculty Leave Approval:** Review and approve leave applications submitted by departmental teachers.

5. Faculty Mentor — *mentor.anil@apex.edu*

Primary Mandate: Student Cohort Advisor & Counselor

- **Assigned Cohort Monitoring:** Direct academic and attendance tracking for an assigned cohort of 20–30 students.
- **AI Early Warning Radar:** View algorithmic risk warnings for students falling behind in grades or attendance.
- **Mentorship Communication:** Dedicated private chat threads with mentees and parent contact logs.
- **Student Leave Recommendations:** First-tier approval or recommendation on student medical and personal leave requests.

6. Faculty Instructor — faculty.priya@apex.edu

Primary Mandate: *Classroom Instructor & Evaluator*

- **Lecture & Lab Attendance Marking:** Mark session attendance with 1-click bulk present and individual absent toggles.
- **Assignment Lifecycle:** Create assignments, upload reference files, set submission deadlines, and evaluate student uploads.
- **Exam Marks Entry:** Enter internal and examination scores with real-time grade calculations.
- **Personal Timetable & Timesheets:** View weekly class schedule and record biometric check-in/out timestamps.

7. Accountant / Bursar — accountant.raman@apex.edu

Primary Mandate: *Finance, Billing & Payroll Officer*

- **Fee Structure Management:** Configure course fee items (tuition, laboratory, library, developmental).
- **Student Billing & Invoicing:** Issue term invoices, apply concessions, record manual/gateway payments, and issue receipts.
- **Defaulter Collections:** Track unpaid student fee balances and trigger automated fee reminders.
- **Staff Payroll Generation:** Run monthly faculty salary calculations, deductions, and issue confidential payslips.

8. Student — student.rahul@apex.edu

Primary Mandate: *Enrolled Academic Scholar*

- **Personal Academic Portal:** Live weekly timetable, subject directory, syllabus outlines, and attendance status.
- **Attendance Radar:** Real-time subject-wise percentage with statutory 75% safety margin indicator.
- **Assignment Submissions:** View pending deadlines, download problem sets, and upload digital submissions.
- **Exam Results:** View published semester report cards and internal assessment scores.
- **Fee Invoices & Receipts:** Pay tuition bills via simulated gateway and download instant PDF receipts.
- **Print Shop & Library:** Order campus print jobs (retrieve 4-digit PIN) and view borrowed library books.

9. Parent / Guardian — parent@apex.edu

Primary Mandate: *Student Guardian & Sponsor*

- **Ward Academic Oversight:** View child's live attendance percentage and statutory risk warnings.
- **Grade Reports:** Review published mid-term and semester examination scorecards.
- **Fee Settlement:** Monitor pending semester dues and settle outstanding invoices directly.
- **Official Communications:** Receive institutional circulars, emergency announcements, and mentor updates.

10. Campus Security Officer — security.chief@apex.edu

Primary Mandate: *Campus Safety & Gate Operations*

- **CCTV Zone Health:** Live monitoring of camera operational status across campus gates, blocks, and perimeter.
- **Student Sighting Radar:** Real-time edge computer vision detection feed displaying student entry and zone timestamps.
- **Student Journey Stepper:** View chronological breadcrumb trail of student campus sightings.
- **Audit-Logged Inquiries:** Search student locations with mandatory justification logging for regulatory privacy compliance.

11. Print Shop Staff — printstaff.dev@apex.edu

Primary Mandate: *Campus Fast-Print Kiosk Manager*

- **Print Queue Fulfillment:** Live queue of submitted print orders across the student and faculty body.
- **4-Digit PIN Validation:** Secure verification of pickup PINs before releasing printed documents.
- **Cost & Inventory Accounting:** Page count tallies, color vs. monochrome accounting, and revenue tracking.

12. Library Staff — *librarystaff.anita@apex.edu*

Primary Mandate: *Librarian & Resource Manager*

- **Catalog Management:** Add and categorize books, authors, ISBNs, and physical shelf/rack allocations.
- **Circulation Desk:** Process book issues and returns via student barcode/ID lookup.
- **Overdue Loan Management:** Automatic ■5/day fine calculations, fine waivers, and lost book processing.

4. Production Operations, Hardware Integration & Compliance

Guidelines for operating CampyTeq in a live university campus environment with hardware peripherals and privacy compliance.

Hardware Peripheral Interfacing

- **RTSP IP Cameras (CCTV):** Deploy IP cameras supporting RTSP streaming (H.264/H.265). Stream URLs are registered in `cv-service/stream_simulator.py` using OpenCV/YOLO video capture feeds. Sightings are sent via authenticated HTTP POST to `/api/v1/tracking/sightings/`.
- **Biometric Scanners (ZKTeco, eSSL):** Connect local biometric attendance terminals via LAN push SDK. Fingerprint/face scans trigger automated attendance logging against `/api/v1/attendance/faculty-checkin/` or classroom rosters.
- **Print Kiosks & Network Printers:** Print shop terminals connect to campus multi-function printers using the **CUPS** (Common Unix Printing System) or **IPP** (Internet Printing Protocol). Jobs are spooled upon 4-digit PIN verification.

Production Cloud & Database Topologies

- **Containerized Orchestration:** Multi-container stack managed via `docker-compose.prod.yml` comprising Nginx (Reverse Proxy), Django (Gunicorn WSGI), Next.js (Node 20 Cluster), PostgreSQL 16 (Relational DB), and Redis 7 (Cache/Queue).
- **Automated Database Backups:** Nightly compressed PostgreSQL dumps (`pg_dump | gzip`) scheduled via cron, stored in off-site encrypted object storage (AWS S3 / Cloudflare R2) with 14-day rolling retention.
- **Zero-Downtime Migration Routine:** Run `python manage.py migrate --noinput` within transient Docker execution containers prior to rolling container image updates.

Security, Ethical AI & Data Privacy Compliance

- **Zero Student GPS Tracking:** CampyTeq explicitly rejects continuous background GPS tracking in the mobile application. Student whereabouts are strictly logged via authorized physical campus CCTV checkpoints.
- **Data Retention & 30-Day Auto-Purge:** In compliance with the Digital Personal Data Protection (DPDP) Act, FERPA, and GDPR, raw CCTV student sightings and biometric camera embeddings are automatically purged after 30 days via scheduled database maintenance routines.
- **Ethical Decision-Support Guardrail:** The AI Academic Early Warning Radar operates strictly as an advisory tool for human faculty mentors. Automated punitive disciplinary actions or grade decrements are prohibited by system design.
- **Rate Limiting Safeguards:** Nginx reverse proxy enforces strict request rate ceilings (5 requests/sec on authentication endpoints; 30 requests/sec on general REST API routes) to neutralize brute-force credential stuffing and DDoS attempts.

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